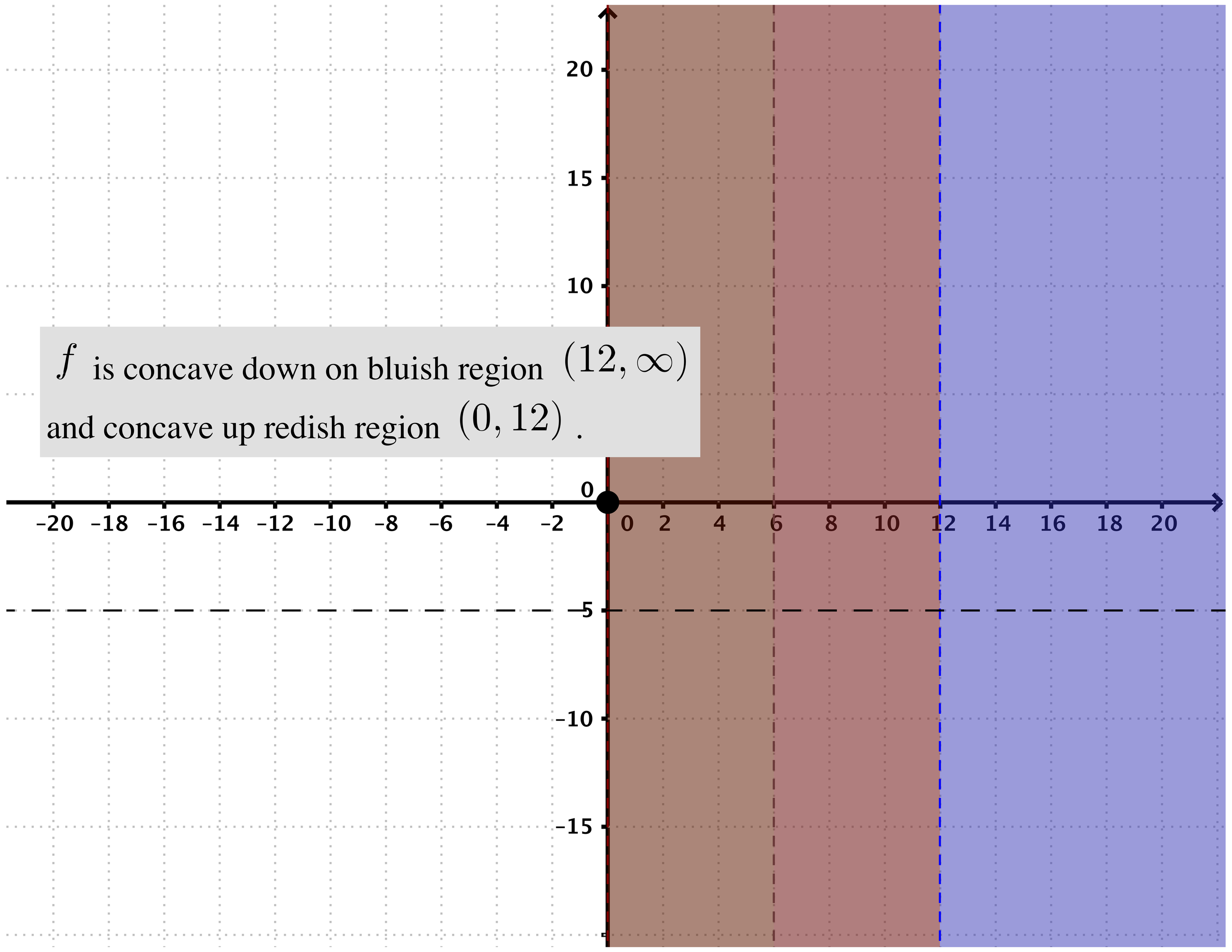

f is concave down on bluish region $(12, \infty)$ and concave up redish region $(0, 12)$.